

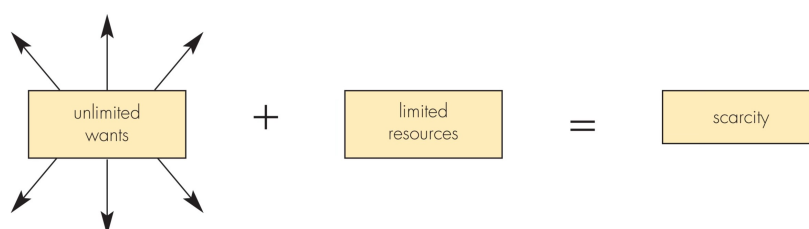
AS Level 01

Basic economic ideas and resource allocation

1.1 Scarcity, choice and opportunity cost

○ The fundamental economic problem and scarcity.

Scarcity refers to the fact that at any moment in time, the output that an economy is able to produce will be limited by the resources and technology available. People's **wants** and **needs**, however, will always exceed the **resources** available to satisfy them — in other words, these wants and needs are unlimited. This is known as the fundamental **economic problem**.



▲ **Figure 1.1** The real cause of the economic problem

KEY TERMS

wants: items that are not essential for survival (e.g. a new car or television)

needs: items that are essential for survival (e.g. food or shelter)

resources: the inputs available to an economy for use in the production of goods and services

economic problem: a situation where there are not enough resources to satisfy all human needs and wants

opportunity cost: the benefit forgone from not choosing the next best alternative

» As a result of this condition of scarcity, choices must be made.

» In all economies, therefore, there is an inevitability of choice at all levels of decision making — at the level of the individual, the firm and the government.

This focus on choice stresses the need to recognise the implications not only of choosing one thing, but also of not choosing something else. **Opportunity cost** is the benefit forgone from not choosing the next best alternative. An example is using a piece of land for farming purposes rather than building a factory on it.

As a result of the condition of scarcity, choices must be made. This could be at the level of:

» individuals, firms, governments

○ The nature and definition of opportunity cost, arising from choices.

The focus on choice stresses the need to recognise the implications not only of choosing one thing, but also of not choosing something else. This is known as opportunity cost.

KEY CONCEPT

Scarcity and choice: the fundamental problem in economics is that resources are scarce and wants are unlimited, so it is always necessary to make a choice between competing uses for the resources and there is always an opportunity cost in making this choice.

○ The basic questions of resource allocation

The emphasis on choice focuses on three basic economic questions:

» **what to produce**

» **how to produce**

» **for whom to produce**

The three basic economic questions are solved in different ways in various economies — in other words, resource allocation can be approached through different systems or mechanisms, as section 1.4 of this chapter shows.

KEY SKILL

Problem solving: understanding the potential missed opportunities forgone by economic agents when choosing one policy over another allows for better decision making.

▶ EXAM-STYLE QUESTIONS: MULTIPLE CHOICE

- 1 The fundamental economic problem exists because:
 - A resources are scarce in relation to unlimited wants.
 - B a business does not know how much it can sell.
 - C resources are scarce in relation to needs.
 - D we cannot always have what we want.
- 2 Wants are unlimited because:
 - A there will always be something we would like to have.
 - B there will always be something we need.
 - C our income is limited.
 - D we are never aware of what we might want in the future.
- 3 A government has limited resources to fund the wants of its population. This means that:
 - A the government has to make choices.
 - B the government is unable to provide any goods free of charge.
 - C the government has to cut the pay of government workers.
 - D the government is unable to borrow more funds.
- 4 A teacher usually marks examination papers during the school break. She receives \$900. Instead, this time, she has decided to do some maintenance work on her apartment. To get someone to do the work would cost \$600. What is the opportunity cost if the teacher decides to do the maintenance work herself rather than mark examination papers?

A \$300. B \$600. C \$900. D \$1500
- 5 You are a student preparing for an assessment. A friend calls round and asks you to play a game of football. You decide to stay at home to prepare your assessment. What is the opportunity cost?
 - A Your friend's team loses the football match.
 - B You save money on bus fares by not playing.
 - C The healthy exercise you will get by playing football.
 - D A higher mark for your assessment.
- 6 Which of these questions is not one an economy has to address?
 - A how resources are used
 - B for what resources are used
 - C where resources are used
 - D for whom resources are used

KEY TERMS

positive statement: a statement that is factual and objective

normative statement: a statement that is subjective and expresses a value judgement

value judgement: an opinion that reflects a particular point of view

1.2 Economic methodology

○ Economics as a social science

A social science can be defined as the scientific study of human society. Can economics be regarded as a social science?

- » Economics is social in the sense that it studies different aspects of human behaviour and, in particular, the choices that humans make.
- » Economics is a science in the sense that it uses an organised system of theories and facts capable of making verifiable predictions.
- » Economics can therefore be regarded as a social science because it uses scientific methods to establish theories that can help explain the behaviour of individuals, groups and organisations in societies.

○ Positive and normative statements

It is important in economics to be able to distinguish between two different types of statements — positive statements and normative statements.

- » A **positive statement** is one that can be checked against the facts to decide whether it is true.
- » A **normative statement**, on the other hand, reflects the norms or values of the person expressing the statement — such a statement will involve a **value judgement** and will reflect someone's personal opinions. Normative statements often include the words 'should' or 'ought to'. The distinction between facts and value judgements is therefore very important in economics.

○ The meaning of the term 'ceteris paribus'

- » Although economics is one of the social sciences, with many aspects of the subject involving scientific analysis, it is not really possible to study human behaviour under laboratory conditions.
- » However, economic theory does assume that certain aspects of human behaviour can be held constant.
- » This assumption of **ceteris paribus**, that other things are equal, means that economists can analyse one aspect of human behaviour at a time. For example, in this way it has been possible to put forward **economic laws** of demand and supply. These economic theories have been put forward in relation to both **microeconomics** and **macroeconomics**.

○ The importance of the time period

Economists, when analysing economic behaviour, distinguish between three different time periods:

- » **Short run**: this refers to that time period in which only certain factors of production (factors of production are covered in section 1.3 of this chapter) can change. These are known as 'variable factors'. In the short run it is not possible to change the 'fixed factors'. For example, in the short run it may be possible to change labour, but the same capital will need to be used.
- » **Long run**: this refers to that time period when the inputs of all factors of production can be changed — for example, it will be possible to vary both labour and capital in the long run. It is not possible to define exactly how long the short run or the long run is because it will vary depending on the particular circumstances.
- » **Very long run**: this refers to that time period when supply conditions can change because of technical progress. In both the short run and the long run, technical progress is assumed to be held constant. In the very long run, however, technical progress can change — for example, as a result of a new invention in a particular industry — and this will have an effect on the supply conditions in that industry.

KEY TERMS

ceteris paribus: a Latin term that literally means 'other things being equal'

economic law: an economic theory put forward by economists (e.g. the laws of demand and supply)

microeconomics: the study of the behaviour of relatively small economic units (e.g. particular individuals, households or firms)

macroeconomics: the study of economics at the national and international levels

KEY TERMS

short run: the time period when it is not possible to change all of the factors of production

long run: the time period when it becomes possible to change all of the factors of production

very long run: the time period when technical progress is no longer assumed to be constant, as is the case in the short run and the long run, and the conditions of supply in an industry can be affected, e.g. by the impact of a new invention

► EXAM-STYLE QUESTIONS: MULTIPLE CHOICE

1 Which of these is a positive statement?

- A Wages in Indonesia are likely to be unevenly distributed.
- B In 2019, the average wage in Indonesia was 3.8 million Indonesian Rupiah per month.
- C The average hourly wage rate in Jakarta should be higher than elsewhere in Indonesia.
- D The average wage in Indonesia depends on how it is measured.

2 Insert the correct words into this sentence.

A ... (1) ... statement in economics is one that is subjective and ... (2) ... be tested by using real world evidence.

	1	2
A	normative	cannot

	1	2
B	normative	can
C	positive	cannot
D	positive	can

1.3 Factors of production

○ The nature and definition of factors of production

Production in an economy can take place in three sectors:

» **Primary sector:** this is the extractive sector, where minerals are taken from the ground, and is concerned with production in areas of an economy such as farming, fishing, forestry, mining and quarrying.

» **Secondary sector:** this is the manufacturing and construction sector, working with the resources that have been extracted in the primary sector, and is concerned with areas of an economy such as car production and the construction of airport runways.

» **Tertiary sector:** this is the services sector and is concerned with wide areas of economic activity such as banking, insurance, tourism, teaching, medicine and the law.

There are four factors of production:

» **Land:** this refers to all the natural resources that can be used in the process of production. It can include farmland, forests, lakes and rivers and all the mineral deposits of a country, such as coal or oil.

» **Labour:** this refers to all the human input into the process of production. It refers not just to the people themselves, but to their skills, training, education and qualifications. It can also be referred to as 'human capital' or 'intellectual capital'.

» **Physical capital:** this refers to the human-made aids that can be used in the process of production. It can refer to equipment, machinery and factories.

» **Enterprise:** this refers to the factor that brings the other factors of production together to produce products. The individual who combines the other factors of production, and takes a risk in doing so, is an entrepreneur.

○ The difference between human capital and physical capital

» **Human capital** refers to the human component of production — that is, the talent, knowledge, abilities, training, education and skills of the labour force.

» **Physical capital** refers to the non-human resources used in the production of goods and services — for example, the tools, equipment, plant, buildings and machinery.

○ The rewards to factors of production

The rewards to the factors of production are as follows:

» **Rent:** the reward to land.

» **Wages or salaries:** the reward to labour.

» **Interest:** the reward to capital.

» **Profit:** the reward to enterprise; many enterprises aim for profit maximisation.

KEY TERMS

human capital: the skills, knowledge and experience possessed by a population in terms of their value or cost to a business or an economy

physical capital: the tangible, human-made objects that a business uses to produce goods and services (e.g. tools, machinery and equipment)

KEY TERMS

rent: the price paid for the use of land

wage: the reward to labour based on the number of hours worked multiplied by an hourly rate of pay

salary: the reward to labour on an annual basis

interest: the reward for parting with liquidity; the reward to capital for the use of the human-made aids to production

profit: the reward to enterprise, defined as the difference between total revenue and total costs

○ Division of labour and specialisation

Specialisation refers to a process of concentration on a particular aspect of production:

» A car assembly line is a good example of the way in which a manufacturing process can be broken down into a sequence of specific tasks. Workers will concentrate on, or specialise in, these particular tasks, giving rise to a **division of labour**.

» One of the first studies of this process was by the Scottish economist **Adam Smith**, who described in his book **The Wealth of Nations (1776)** how division of labour in a pin factory enabled a great many more pins to be produced than if each worker tried to do everything him- or herself.

The concept of specialisation is also discussed in the context of international trade in Chapter 6, section 6.1.

KEY TERMS

specialisation: the process whereby individuals, firms and economies concentrate on producing those products in which they have an advantage

division of labour: the way in which production is divided into a sequence of specific tasks which enables workers to specialise in a particular type of job

Adam Smith: one of the founding fathers of economics (1723–90) and author of *The Wealth of Nations*, published in 1776

○ The role of the entrepreneur in contemporary economies

Entrepreneurs play a crucial role in contemporary economies, performing two key functions:

» **Organisation:** entrepreneurs are responsible for organising and coordinating the other factors of production — land, labour and capital — to produce goods and services.

» **Risk:** entrepreneurs take a risk in performing this organisation and coordination function; this arises from the uncertainty that will be a feature of any initiative they take. Although there are many famous entrepreneurs in the world, who have had success in a number of different business ventures, there are many others who have failed.

Contemporary economies have provided many opportunities for the development of an enterprise culture. This is where people are imaginative and creative, and are willing to take risks in order to gain profit.

There are a number of ways in which a government could encourage the development of an **enterprise culture**, including:

- » supporting business start-up programmes
- » encouraging venture capital financing (this is where private investors provide finance to start-up businesses that are believed to have good long-term growth potential)
- » providing grants to support research and development
- » policies to promote competition in markets, such as deregulation (this is covered in Chapter 8, section 8.1)
- » development of appropriate education and training to improve the quality of human capital
- » financial support for the development of technology parks and the fostering of innovation
- » favourable tax treatment for start-up businesses in the form of tax incentives
- » reducing administrative burdens, such as less 'red tape' in the form of excessive bureaucratic paperwork

► EXAM-STYLE QUESTIONS: MULTIPLE CHOICE

1 What are the factors of production?

- A natural resources that are available for production
- B another term for resources in an economy
- C what are required in the future to produce goods and services
- D the human resources used in the production of goods and services

2 The factor of production capital means:

- A what is combined with land and labour to produce goods and services.
- B the value of assets owned by a firm.
- C money needed to set up a firm.
- D money provided by shareholders to expand production in a firm.

3 Which of these is the best explanation of how an economist describes an entrepreneur?

- A someone who is lucky and makes a lot of money
- B someone who succeeds in some but not all business activities
- C someone who is in charge of a firm
- D someone who seeks out business opportunities and is willing to take risks

4 Which of these is an example of land as a factor of production?

- A an airline pilot
- B an airfield
- C an aeroplane
- D aircraft fuel

5 Specialisation can only occur when:

- A a price has been agreed to buy and sell a product.
- B a firm has modern methods of production.
- C a firm is more efficient than all others in making a product.
- D an entrepreneur sees a business opportunity.

6 Three workers (A, B and C) are employed in a shoe factory. The table below shows the output of each worker if they divide their time equally between the three stages in the production process.

Soles Uppers Finishing and polishing

Worker A 100 Worker B 40 Worker C 40

40 40 100 40 40 100

Each worker is now asked to work on the part of the production process where they are a specialist. What would be the increase in the total production of shoes?

- A 140 shoes
- B 180 shoes
- C 360 shoes
- D 540 shoes

7 Which reason is most likely to explain why an entrepreneur is prepared to take risks?

- A to secure a large bank loan
- B to realise a business ambition
- C to justify giving up a safe job
- D to increase their self-esteem

1.4 Resource allocation in different economic systems

○ Decision making and resource allocation in market, planned and mixed economies

An **allocative mechanism** is needed for deciding how economic goods (see section 1.6 of this chapter) that are scarce are produced and consumed.

There are three different types of allocative mechanism:

» market economies, planned economies, mixed economies

○ Market economies

In a **market economy**, the allocation of resources is left to the **market** forces of demand and supply, operating through the price mechanism. The advantages and disadvantages of the market economy are shown in Table 1.1.

▼ Table 1.1 Advantages and disadvantages of the market economy

Advantages of the market economy	Disadvantages of the market economy
- Decisions are made by individual consumers, who act in their own self-interest, i.e. seek to maximise their utility or satisfaction when they consume a product. - Decisions are made by individual producers, who act in their own self-interest, i.e. seek to maximise their profits. - The use of the price mechanism to allocate resources (referred to as 'the invisible hand' by the Scottish economist Adam Smith) means that there is no need for any government intervention in the allocation of resources. - Competition between firms can lead to greater efficiency.	- Some products will be underprovided and underconsumed in a market economy; these are known as merit goods (e.g. education and healthcare). Merit goods are covered in section 1.6 of this chapter. - Some products will be overprovided and overconsumed in a market economy; these are known as demerit goods (e.g. alcohol and tobacco). Demerit goods are covered in section 1.6 of this chapter. - Some products will not be provided or consumed at all in a market economy because it would be impossible to charge a market price for them; these are known as public goods (e.g. defence and lighthouses). Public goods are covered in section 1.6 of this chapter. - Income and wealth disparities can be very significant.

○ Planned economies

Planned economies, also known as **command economies**, involve the allocation of scarce resources through government intervention with no (or very little) scope for market forces to operate. The advantages and disadvantages of planned economies are shown in Table 1.2.

Advantages of the planned economy	Disadvantages of the planned economy
- Government intervention in the allocation of resources means it can take decisions in the national interest (e.g. it can prevent the production of socially undesirable products such as drugs). - The government can intervene to bring about a more equitable distribution of income and wealth.	- A system with such a large amount of government influence and control will tend to be bureaucratic and, as a result, may be inefficient. - The lack of competition and the lack of the profit motive mean that products are often of poor quality with consumers having little choice.

○ Mixed economies
A **mixed economy** combines elements of both market economies and

planned economies — in other words, there is some degree of state ownership and state intervention, but in many areas of the economy market forces will be allowed to operate.

It could be argued that all economies today are, to some extent, mixed economies. However, there are large differences between, say, China, where the government still plays an important role in the allocation of resources, and the USA, where the government has only a limited role in the allocation of resources.

○ Transitional economic systems

A number of economies are going through a period of change where the extent of central planning is being reduced and market forces are being allowed to have a greater degree of influence. China and Cuba are examples of such a **transitional economy**.

There are, however, possible problems associated with transition, as Table 1.3 shows.

▼ Table 1.3 Problems of transitional economies

Unemployment	A planned economy is generally better able to keep down the rate of unemployment in an economy; when there is a move towards greater reliance on market forces, the rate of unemployment in an economy is likely to increase because, in a market economy, firms aim to maximise profits and this may lead them to reduce costs of production, possibly by laying off some workers.
Inflation	In a planned economy, the state controls prices so it is easier to keep down the rate of inflation; when prices are determined by the free-market forces of demand and supply, it is more difficult to control prices and so inflation is more likely.
Output	In a planned economy, it is possible for the state to support inefficient firms and industries; when state support is ended, such firms and industries may not be able to compete and so output could fall.
Welfare	A planned economy is able to provide housing and healthcare to everyone; with the introduction of market forces, there may be a fall in welfare provision and this may have a detrimental effect on levels of productivity in the economy.

► EXAM-STYLE QUESTIONS: MULTIPLE CHOICE

1 Which of these terms describes a country where government planning, regulations and directives are responsible for the allocation of resources?

- A capitalist economy
- B mixed economy
- C market economy
- D planned economy

2 A criticism of a planned economy is that it puts more emphasis on producing capital goods than producing consumer goods. This is because:

- A the system of decision-making is too centralised.
- B there are shortages of many consumer goods.
- C the government's focus is on growth and not current consumption.
- D consumers cannot afford to buy any more consumer goods.

3 A planned economy has recently become a mixed economy. This process has involved a change in the composition of what it produces. What is the most likely consequence of this change?

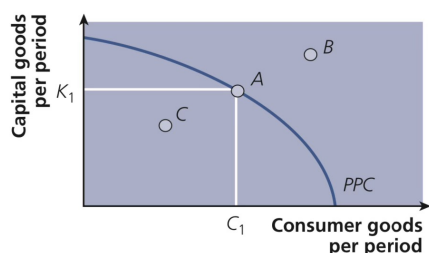
- A an increase in food prices

- B an increase in imports
 C an increase in structural unemployment
 D an increase in the growth rate of the economy
- 4 A mixed economy can best be described as one in which:
 A production is determined by the central government and the market mechanism.
 B agricultural and manufactured goods are produced.
 C production is determined by entrepreneurs.
 D consumer goods and capital goods are produced.
- 5 Which of these is the most likely outcome when a mixed economy privatises its rail services?
 A Peak train fares will increase.
 B Off-peak train fares will decrease.
 C Fewer rail services will be provided.
 D Train fares will now be determined by supply and demand.
- 6 After a planned economy changed to a more market economy, there was a reduction in the proportion of household income that was being saved. How might this be explained?
 A Households were more concerned about the economy's future economic prospects.
 B More 'big ticket' goods had become available.
 C New charges were introduced for secondary education.
 D Household incomes fell.

1.5 Production possibility curves

- The nature and meaning of a production possibility curve (PPC)
 A **production possibility curve** (or **production possibility frontier**, as it is sometimes called) shows the different combinations of products that can be produced if an economy is working at full capacity.
- The shape of the curve: constant and increasing opportunity costs
 The shape of the curve shows that there are a number of different combinations of products that can be produced. It is drawn as a curve rather than as a straight line because not all factors of production are equally efficient.

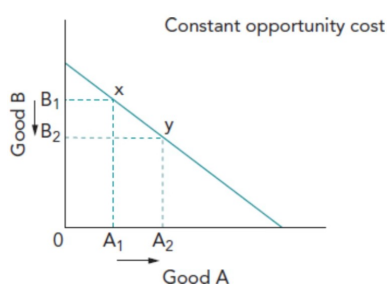
» The production possibility curve (PPC) in Figure 1.1 shows the combination of capital goods (shown on the vertical axis) and consumer goods (shown on the horizontal axis) that an economy can produce in a particular period of time with the existing economic resources available.



▲ Figure 1.1 A production possibility curve

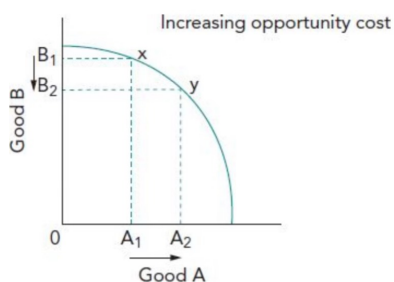
- » Point A shows one possible combination of outputs, where the economy produces K_1 capital goods and C_1 consumer goods.
- » Any movement along the curve from point A shows that the production of more of one type of good leads to the production of less of the other (thus illustrating the concept of opportunity cost).
- » Point C, which is inside the PPC, shows that the economy is not using its resources efficiently and there is some unemployment of resources. Output of both capital and consumer goods is lower than it could be.

- Constant and increasing opportunity costs.



- » It has already been stated that a production possibility frontier is drawn as a curve, rather than as a straight line, because not all factors of production are equally efficient.
- » It is therefore necessary to distinguish between constant and increasing opportunity costs. If it were possible to move from one point on the production possibility curve to another, with an equal sacrifice of

resources, then this would indicate a situation of constant opportunity costs.

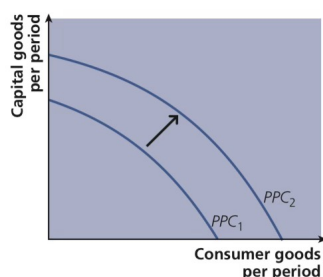


» However, there will come a time when this is not the case. Increasing opportunity costs mean that an ever-increasing amount of one product will need to be sacrificed to produce more of the other product.

» The reason is that different factors of production have different qualities. As a result of this, the production possibility frontier changes shape slightly as it approaches each axis.

» For example, in Figure 1.1, this is most clearly seen as the PPC gets closer to the horizontal axis, showing the consumer goods produced per period of time.

○ The causes and consequences of shifts in a PPC.



» Point B in Figure 1.1, which is outside the *PPC*, is unreachable at the present time given the resources that the economy currently has.

» However, over a period of time it is possible for there to be **economic growth** resulting from the availability of more resources and/or the more productive use of resources, and this would enable point B to be reached. This can be seen in Figure 1.2.

Economic growth (see Chapter 9, section 9.2, on the distinction between actual economic growth and potential economic growth) enables an economy to produce more of both capital and consumer goods. It refers to a situation where there is an expansion in the productive capacity or

potential output of an economy. This is shown in Figure 1.2 by a rightward shift of the *PPC* from *PPC1* to *PPC2*.

Of course, if there were a decrease in the quantity and/or quality of resources in an economy, this would lead to a leftward shift of a *PPC* from *PPC2* to *PPC1*.

○ The significance of a position within a *PPC*.

» It is important to understand the significance of a position within a *PPC*.

» As can be seen in Figure 1.1, point A is where an economy is using its resources efficiently, but point C, within a *PPC*, is where an economy is using its resources inefficiently.

» At this point, not all resources are being utilised and the output of both products is lower than it could be if all resources were being used.

► EXAM-STYLE QUESTIONS: MULTIPLE CHOICE

1 A production possibility curve is drawn on the assumption that:

- A there are always opportunities to produce more of one good.
- B all existing resources are being used and there is a given state of technology.
- C all existing resources are being used.
- D production never takes place within or outside the production possibility curve.

2 A country produces at a new point within its production possibility curve when:

- A a new technology is being applied to increase production.
- B there is an increase in capital investment by firms.
- C more is needed of one good and not the other good.
- D there is a fall in unemployment.

3 An economy produces two goods: cars and smartphones. Why is the production possibility frontier curved outwards from the origin?

- A All resources are being fully used.

B The opportunity cost of producing cars or smartphones is constant.

C An increase in production of cars can only occur if there is a decrease in the production of smartphones.

D Some of the economy's resources are better suited to the production of cars, other resources better suited to the production of smartphones.

4 An economy produces two goods: cars and smartphones. Its production possibility curve shifts outwards. What would have caused this shift?

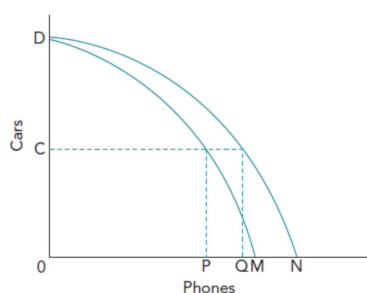
A the discovery of a new valuable oil reserve

B closure of an inefficient car assembly plant

C an increase in sales tax on cars and smartphones

D a fall in the rate of unemployment

5 An economy produces two goods: cars and smartphones. The diagram shows that, as a result of a technological advance, more smartphones can be produced for the same resources.



What is the change in the opportunity cost of producing OC cars?

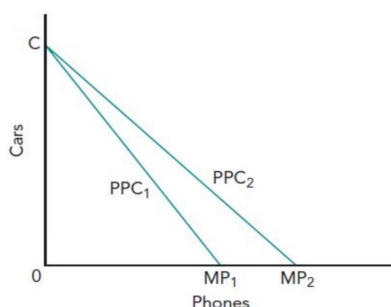
A PN to PM

B PM to QN

C PQ to PM

D QM to PM

6 The diagram shows that the production possibility curve of cars and smartphones has changed from PPC1 to PPC2.



What has happened to the opportunity cost of car production and the returns to factors producing smartphones?

	Opportunity cost of car production	Returns to factors producing smartphones
A	fallen	fallen
B	fallen	risen
C	risen	risen
D	risen	fallen

1.6 Classification of goods and services

○ The nature and definition of free goods and private goods (economic goods).

KEY TERMS

free good: a good that is not scarce and so does not require a market price to be attached to it

private good: a good that is bought and consumed by individuals for their own benefit

rivalry: a feature of private goods whereby when a product is consumed by one person, it cannot be consumed by another

excludability: a feature of private goods whereby people can be excluded from consuming a good

Free goods

A free good is one which is consumed by people without a situation of scarcity arising — in other words, there is enough of the good to satisfy everybody. As the good is not scarce, it does not require a market. The supply of the good equals the demand for it at zero price. It takes no factors of production to produce a free good and so there is no opportunity cost involved.

Private goods

A private good (or economic good) is one which is consumed by an individual for their own private benefit. This applies to most products in an economy. Private goods have two important characteristics:

- » **Excludability**: one key feature of a private good is that people can be excluded from consuming it.
- » **Rivalry**: another key feature of a private good is that the consumption of it by one person reduces its availability for other people; there is rivalry in such a situation because consumers are in competition with other consumers to consume a particular product.

	Excludable	Non-excludable
Rival	Private good	Quasi-public good
Non-rival	Quasi-public good	Public good

○ The nature and definition of public goods.

In contrast to private goods, **public goods** are provided by society as a whole so that everyone can benefit from them.

Public goods have two important characteristics:

- » **Non-excludability**: once a public good has been provided for one person, it is not possible to stop other people from benefiting from such a good (i.e. no one is excluded).
- » **Non-rivalry**: as more people consume the public good, the benefit to those already consuming it is not reduced (i.e. consumption by one person does not prevent others from consuming it).

KEY TERMS

public good: a good that is non-rival, non-excludable and non-rejectable

non-excludability: where the consumption of a product by one person does not exclude others from consuming the same product

non-rivalry: where the consumption of a product does not prevent its consumption by someone else

free rider: the idea that it would be impossible to charge people for using a good or service because it would be impossible to prevent someone who had not paid from benefiting

government expenditure: the total of all spending by a government

non-rejectability: where individuals cannot actually avoid the consumption of a public good, even if they want to

These products need to be provided by the state or the public sector because if they were provided by the private sector, it would be impossible to exclude someone who had not paid. This gives rise to the **free rider** problem.

For example, it would not be possible to provide street lighting through the private sector because it would be impossible to prevent someone who had not paid from benefiting from the service. When such products are provided by the public sector, they are part of **government expenditure** and are financed out of taxation.

In addition to being non-rival and non-excludable, public goods are also **non-rejectable**. This means that, even if a person does not want to be protected by their country's defence and police system, they are not actually able to reject it.

○ The nature and definition of merit goods

KEY TERMS

merit good: a product that is rivalrous and excludable but, if left to a free market, would be likely to be underproduced and underconsumed

information failure: where people lack the full information that would allow them to make the best decisions about consumption

market imperfection: a feature of a market which does not perform perfectly because of a failure to make an optimal use of resources, necessitating government intervention

market failure: a market imperfection which gives rise to an allocation of scarce resources which is not as efficient as it might otherwise have been

» A merit good is a particular type of private good. Examples include education and healthcare.

» Like other private goods, merit goods are both rival and excludable, but what distinguishes a merit good is that there is information failure which means that the good is likely to be underprovided and underconsumed if provided through the private sector.

» For example, people don't fully

appreciate the value of a good education or good health. This could be regarded as a market imperfection.

- » Without government intervention, it is likely that there would be market failure because the allocation of resources would be sub-optimal.
- » Governments therefore intervene by providing such goods through the public sector, alongside private sector provision, so that those who would not or could not afford to consume them in the private sector will do so in the public sector.

○ The nature and definition of demerit goods

- » Demerit goods are the opposite of merit goods. Whereas merit goods would be underprovided and underconsumed in a free market, demerit goods would be overproduced and overconsumed in a free market.
- » A demerit good is socially undesirable in some way: for example, alcohol and tobacco.
- » The overproduction and overconsumption of demerit goods is a result of imperfect information by consumers. For example, they may not realise that alcohol and tobacco are bad for their health.
- » Without government intervention, it is likely that there would be market failure because the allocation of resources would be sub-optimal.

► **EXAM-STYLE QUESTIONS: MULTIPLE CHOICE**

1 Which of these is a distinctive characteristic of a pure public good?

- A Consumption by one person means that there is less available for another person.
- B Nobody can be excluded from consuming it.
- C It is produced by the government.
- D It is free of charge at the point of use or consumption.

2 Which of these is a distinctive characteristic of a merit good?

- A It is overprovided in the market.
- B It is underprovided in the market.
- C It is funded by the government.
- D It is only given to those who deserve it.

3 Which of these is a distinctive characteristic of a demerit good?

- A It is overprovided by the market.
- B It is underprovided by the market.
- C It is paid for by individuals.
- D Anyone can buy it.

4 Suppose a public museum charges an entry fee. Which of the following applies to visits to the museum when there are very few visitors?

	Excludable	Rival
A	no	no
B	no	yes
C	yes	no
D	yes	yes

5 Which of these is most likely to be a merit good?

- A a local public library
- B cigarettes
- C sea defences
- D satellite television

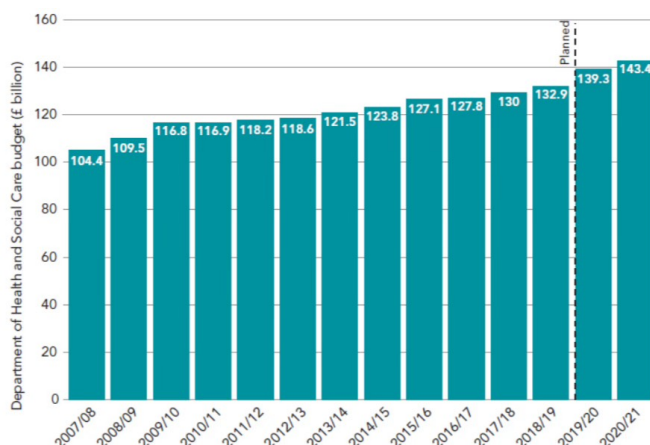
6 The vaccination of a child against polio is a merit good because:

- A the child's family cannot afford the vaccination.
- B it protects the child from polio.
- C it is underprovided by the market because of information failure.
- D it provides employment in the pharmaceutical industry.

► **EXAM-STYLE QUESTIONS: DATA RESPONSE AND ESSAY**

• **Data response**

1. Difficult choices for the UK's National Health Service (NHS)



The core principle of the UK's National Health Service (NHS) is that, with a few exceptions, treatments are available to all UK residents free of charge at the point of use. Funding for the NHS, like other areas of government spending, comes directly from taxation. Figure 1 shows the extent of this funding from 2007/08 to 2018/19 and projected spending for 2019/21.

Figure 1: UK expenditure on health and social care 2007/08 to 2020/21 (real terms at June 2019 prices)

The NHS budget is under increasing pressures. People are living longer, new medicines and treatments are expensive and the number of people eligible for NHS treatment has increased.

It is not surprising, therefore, that there is growing support for charges to be made for more healthcare services, especially for people who can afford to pay. Such a move would make these healthcare services more of a private good than a merit good.

a Using the data in Figure 1, describe how expenditure on the NHS has changed since 2007–08..... [2]

b i Explain why the data are in 'real terms at June 2019 prices'?[2]

ii Consider another way by which the data in Figure 1 might be made more informative..... [3]

c Using the concept of opportunity cost, explain why the NHS is increasingly faced with having to make difficult choices. [3]

d Explain two factors of production that are required for the provision of healthcare services.. [4]

e Discuss whether healthcare services should be provided by the government or by the private sector.[6]

• Essays

2. a A middle-income country increases its expenditure on capital goods, with no increase in expenditure on consumer goods.

With the help of a diagram, explain how this affects the country's production possibility curve and consider whether producers or consumers will benefit most from the change.....[8]

b Discuss how the allocation of consumer goods and capital goods differs between a planned economy and a market economy..... [12]

3. a A public good has the characteristics of non- excludable and non-rival.

Explain how these characteristics can be applied to the provision of flood defences and consider which of the characteristics is most important..... [8]

b Discuss the extent to which libraries and museums should be provided by the government or by the private sector..... [12]

SUMMARY

In this chapter you have learned:

- » the fundamental economic problem of scarcity
- » the need to make choices at all levels, including individuals, firms and governments
- » the nature and definition of opportunity cost
- » the three basic questions of resource allocation
- » the idea of economics as a social science
- » to distinguish between positive and normative statements
- » the meaning of the term 'ceteris paribus'
- » the importance of the time period in relation to the short run, long run and very long run
- » the nature and definition of the factors of production: land, labour, capital and enterprise
- » the rewards to the factors of production
- » the meaning of division of labour and specialisation
- » the role of the entrepreneur in contemporary economies
- » how decisions are made in market, planned and mixed economies
- » how resources are allocated in these three economic systems
- » the nature and meaning of a production possibility curve (PPC)
- » the shape of the PPC in relation to constant and increasing opportunity costs
- » the causes and consequences of shifts in a PPC
- » the significance of a position within a PPC
- » the nature and definition of free goods and private (economic) goods
- » the nature and definition of public goods
- » the nature and definition of merit goods
- » the nature and definition of demerit goods

AS Level 02

The price system and the microeconomy

2.1 Demand and supply curves

○ Effective demand

Effective demand refers to that demand which can be supported by having the means to pay. In this situation, consumers must not just want a particular product, but also be willing and able to pay for it.

○ Individual and market demand and supply

• Individual and market demand

» **Demand** is the quantity of a product that consumers are willing and able to buy at a given price in a given time period.

» An individual demand curve shows the quantity of a product that a particular consumer is willing and able to buy at each and every price, *ceteris paribus* (i.e. with all other things unchanged).

» The individual demand curve will slope downwards from left to right, indicating that a consumer is more likely to buy a product at a lower price than at a higher price. This is known as the **law of demand**.

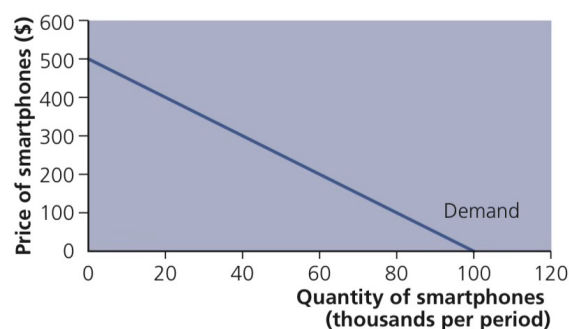
• Aggregation of individual demand curves to give market demand

» A **demand curve** can be drawn for every consumer in a society for every product, but in economics it is more usual to focus on market demand curves.

» Market demand for a product is derived from bringing together (or aggregating) all the potential buyers of a product. It is the total quantity of a product that all potential buyers would choose to buy at a given price in a given period of time.

» A **demand schedule** can be produced for a particular product, such as smartphones.

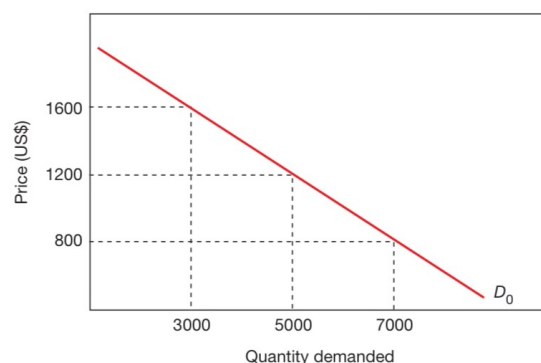
» This schedule can then be plotted to give a market demand curve, as shown in Figure 2.1. The price of smartphones is shown on the vertical axis and the quantity of smartphones bought is shown on the horizontal axis.



▲ Figure 2.1 A demand curve for smartphones

The demand curve shows the relationship between price and the quantity demanded. It is downward sloping, indicating an inverse relationship between the price of a product and the quantity demanded of a product: that is, as the price falls, the demand rises.

Price of a 'standard' PC (US\$)	Quantity demanded per week – demand curve D_0
2,000	1,000
1,800	2,000
1,600	3,000
1,400	4,000
1,200	5,000
1,000	6,000
800	7,000



Derived demand is where the demand for a component depends upon the final demand for a product that uses that component. For example, the demand for rubber is derived from the demand for car tyres. Derived demand can also be used in relation to the demand for workers — for example, the demand for bus drivers derives from people's demand for bus transport.

○ Individual and market supply

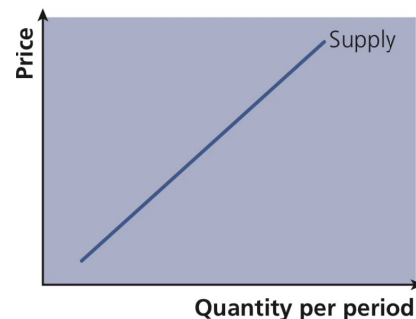
Individual and market supply curves

Supply is the quantity of a particular product that firms are willing and able to sell at each and every price in a given time period, *ceteris paribus* (all other things unchanged). A firm's supply curve will slope upwards from left to right, indicating that a producer will be more likely to sell a product at a higher price than at a lower price. This is known as the **law of supply**.

○ Aggregation of individual firms' supply curves

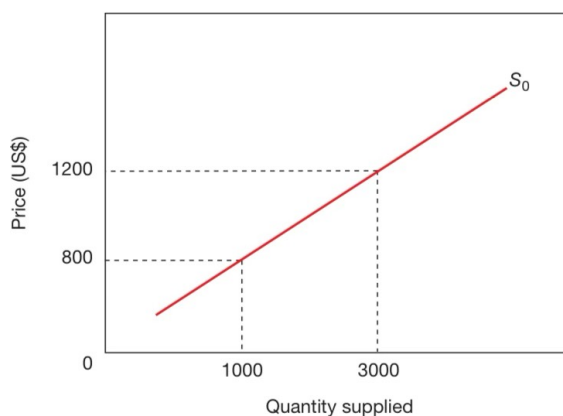
A **supply curve** can be drawn for every producer in an economy for every product, but in economics it is more usual to focus on market supply curves. Market supply of a product is derived from bringing together (or aggregating) all the potential suppliers of a product. It is the total quantity of a product that all potential sellers would choose to sell at a given price in a given period of time.

A supply schedule can be produced for a particular product, such as smartphones. This schedule can then be plotted to give a market supply curve, as shown in Figure 2.2.



▲ Figure 2.2 A supply curve

Price of a 'standard' PC (US\$)	Quantity supplied per week – supply curve S_0
800	1,000
1,000	2,000
1,200	3,000
1,400	4,000
1,600	5,000
1,800	6,000
2,000	7,000



The price of smartphones is shown on the vertical axis and the quantity of smartphones sold is shown on the horizontal axis. The supply curve shows the relationship between price and the quantity supplied. It is upward sloping, indicating a direct relationship between the price of a product and the quantity supplied of a product: that is, as the price rises, the supply rises.

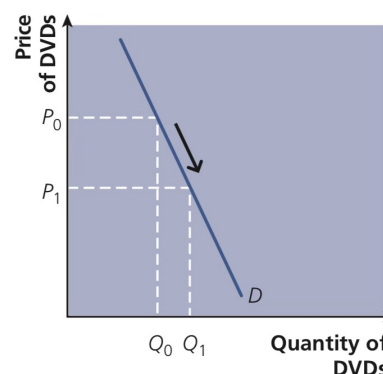
○ The determinants of demand

Price

A major influence on the demand for a product is its price.

Figure 2.3 shows that there is an inverse relationship between a change in the price of a product and the quantity demanded of a product, all other things unchanged (*ceteris paribus*).

When it is only the price of a product that changes, the resulting change in **quantity demanded** can be shown on a demand curve by a movement along the curve. This can be seen in Figure 2.3.



▲ Figure 2.3 A movement along the demand curve

» When the price of a product is reduced, for example, from P_0 to P_1 , the quantity demanded goes up from Q_0 to Q_1 . This is represented by a downward movement along the demand curve, indicated in the diagram by the downwards arrow. This is known as an **extension in demand**.

» If, on the other hand, the price of a product is increased, the quantity demanded falls and this would be shown as an upward movement along the demand curve. This is known as a **contraction in demand**.

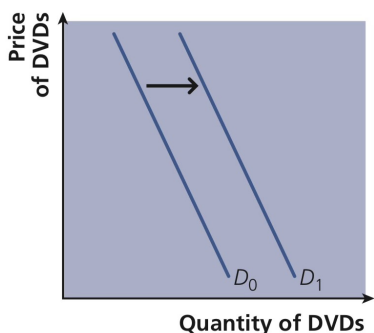
○ The determinants of supply

Movements along a supply curve are determined by changes in the price of a product. Figure 2.4 shows the direct relationship between the price of a product and the quantity supplied of that product. This is why the supply curve is upward sloping. A movement up a supply curve is known as an **extension in supply** and a movement down a supply curve is known as a **contraction in supply**.

○ Causes of a shift in the demand curve

Price is not the only factor that influences demand. If the ceteris paribus assumption is removed, it is possible to consider all the other factors that were previously being held constant. These other factors could include:

- » a change in the incomes of consumers
- » a change in the price of a substitute product (a substitute product is one that could be used for the same purpose by consumers)
- » a change in the price of a complementary product (a complementary product is one that is directly related to, and used with, another product)
- » an advertising campaign
- » a change in population
- » a change in the tastes and preferences of consumers
- » a lowering of interest rates, making borrowing more affordable
- » a change in the weather, possibly associated with different seasons



▲ Figure 2.4 A shift in the demand curve

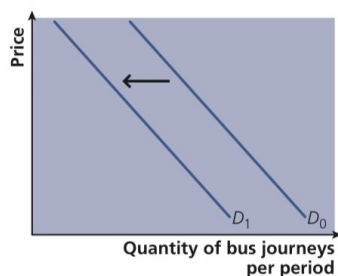
When one of these other factors affects demand, the result is described as a **change in demand** and is shown by a shift of the demand curve. This can be seen in Figure 2.4.

In this diagram, there might have been an increase in incomes and/or an effective advertising campaign. The demand curve shifts to the right, from D_0 to D_1 , as shown by the rightward arrow.

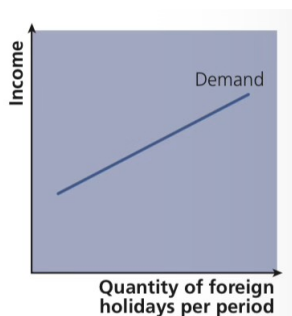
Composite demand refers to the demand for a product that can be used for more than one purpose. Stone, for example, could be used for building purposes and could also be used in the construction of roads; a particular piece of land could be demanded to build both shops and houses.

○ Normal and inferior goods

- » Figure 2.4 showed what usually happens when there is an increase in the incomes of consumers — more of the product is bought at every price and there is a rightward shift of the demand curve, showing an increase in demand. Such goods are called **normal goods**.
- » However, it is possible that the demand for some goods and services decreases when there is an increase in incomes. For example, while there might be an increase in the demand for cars as a result of an increase in incomes, there might be a decrease in the demand for public transport, such as bus journeys.
- » This can be seen in Figure 2.5 where there is a leftward shift in the demand curve for bus journeys, showing a decrease in demand. Such goods are called **inferior goods**.



▲ Figure 2.5 A shift in the demand curve following an increase in consumer incomes (an inferior good)



It is important to recognise that the demand for normal and inferior goods shows the relationship between a change in the quantity demanded and a change in income, not price. Figure 2.6 shows this relationship for a normal good.

▲ Figure 2.6 Demand and income for a normal good

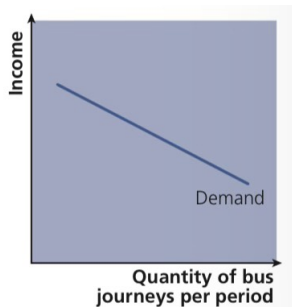


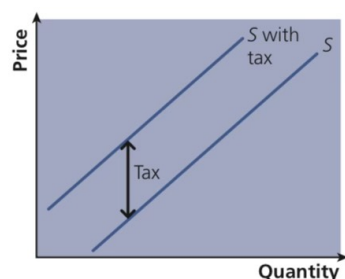
Figure 2.7 shows the relationship for an inferior good.

▲ Figure 2.7 Demand and income for an inferior good

○ Causes of a shift in the supply curve

Price is not the only factor that influences supply. If the ceteris paribus assumption is removed, it is possible to consider all the other factors that were previously being held constant. These other factors could include the following.

Indirect taxes



▲ Figure 2.8 The effect of a sales tax on supply

» A government may decide to impose an **indirect tax**, such as a sales tax, on a particular good or service. Examples are value added tax (VAT) and goods and services tax (GST).

» The effect of the imposition of such a tax can be seen in Figure 2.8. In this case, the tax is a specific tax with a fixed amount of tax per unit, so the supply curve shifts upwards, parallel to the original supply curve.

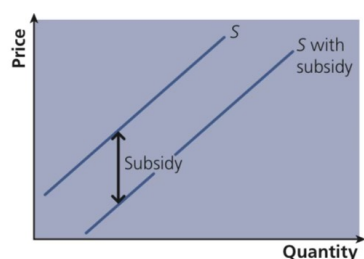
»

Figure 2.8 shows how the imposition of an indirect tax will affect the price of a product. As there is an upward movement of the supply curve, this will lead to an increase in price. In order to determine the exact price charged, it would be necessary to include a demand curve in the diagram (see section 2.4 of this chapter).

○ Subsidies

» The effect of a **subsidy** can be seen in Figure 2.9.

» Whereas the imposition of a tax shifted the supply curve upwards to the left, a subsidy has the opposite effect. If a government pays firms a subsidy to produce a particular

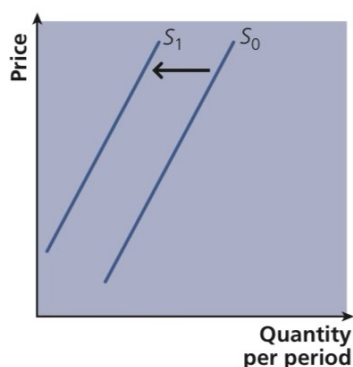


▲ Figure 2.9 The effect of a subsidy on supply

product, this will have the effect of reducing their costs and encourage firms to supply more output at any given price. This can be seen in Figure 2.9 with the supply curve shifting downwards to the right.

The effect of the subsidy, in shifting the supply curve to the right, will be a lowering of price. The actual price will be determined where the 'S with subsidy' line intersects with the demand curve. The effect of the subsidy is that both producers and consumers may benefit (see section 2.4 of this chapter).

○ Production costs

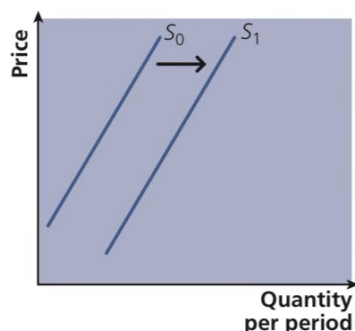


An important influence on supply is the costs of production. If the costs of the inputs in the production process — that is, the costs of the factors of production — increase, then firms will be inclined to supply less output at any given price.

This can be seen in Figure 2.10. The increase in production costs causes the supply curve to shift to the left from S_0 to S_1 . The increase in costs can be seen by the vertical distance between S_0 and S_1 .

▲ Figure 2.10 The supply curve shifts to the left if production costs increase

○ Technology of production



Another important influence on supply is the technology of production. If the technology of production is improved, this means that firms will be able to produce more effectively than before.

This can be seen in Figure 2.11 where improved technology leads to firms supplying a greater output at any given price. The supply curve shifts to the right from S_0 to S_1 .

▲ Figure 2.11 The supply curve shifts to the right if production costs fall

○ The prices of other goods

- » There may be a degree of substitution on the supply side if the prices of different products change.
- » In many cases, the factors of production that a firm has can have alternative uses, so a firm may be influenced by changes in the prices of different products to produce more of one product and less of another.
- » A rise in the price of a product could increase its profitability, so a firm may decide to switch production towards this product. For example, car production companies are increasingly switching production towards electric cars.

○ Expected prices

A final influence on market supply relates to the expectations of firms about possible future prices. This is especially the case in those situations where the production process takes quite a long time. Firms will thus need to take supply decisions on the basis of expected prices in the future. This is often the case in agriculture.

○ The distinction between the shift in the demand or supply curve and the movement along these curves

- » It has already been pointed out that it is important to distinguish between a movement along a demand curve and a shift of a demand curve.
- » If there is a change in the market price of a product, and nothing else changes (i.e. assuming *ceteris paribus*), this will involve a movement along a demand curve. This shows how consumers react to a change in the price of a product. This can be seen in Figure 2.3.
- » If the situation of *ceteris paribus* cannot be assumed, however, and there is a change in any of the other influences on demand, then the demand curve will shift. This could involve a shift to the left, showing a decrease in demand (e.g. as a result of a lowering of incomes) or a shift to the right, showing an increase in demand (e.g. as a result of an effective advertising campaign). The latter case was shown in Figure 2.4.
- » It is also important to distinguish between a movement along a supply curve and a shift of a supply curve. If there is a change in the market price of a product, and nothing else changes

(again assuming *ceteris paribus*), this involves a movement along a supply curve. This shows how firms react to a change in the price of a product and was shown in Figure 2.2.

» If the situation of *ceteris paribus* cannot be assumed, however, and there is a change in any of the other possible influences on supply, then the supply curve will shift because this will affect the willingness of firms to supply at any given price.

» As has already been indicated, this could involve a shift to the left: for example, as a result of the imposition of an indirect tax on the consumption of a product (see Figure 2.8) or an increase in production costs (see Figure 2.10). Alternatively, it could involve a shift to the right: for example, as a result of the introduction of a subsidy (see Figure 2.9) or an improvement in technology (see Figure 2.11).